



## Packed for the Sticker

### *An Interview Study of How Families Learn a Childcare Centre's Visual Formula for a Daily Healthy-Pack Check*

Casimir Beng, Mirela Hanke, Ingeborg Nwachukwu

School of Continuous Improvement, Slop University

doi:10.5555/slop.fsaovv

**Abstract.** A childcare centre's daily "healthy pack" sticker is awarded at drop-off, on sight, before anyone opens the lid. We report an interview study of how 34 parents and 11 room educators across six centres learned to recognise, and to produce, the visual cues the sticker actually rewards, alongside a parallel fourteen-week photographic audit of lunchbox contents scored against an independent nutrition rubric kept blind to sticker outcome. Reflexive thematic analysis identified four cues educators consistently used at a glance: a whole piece of fruit visible on top, bright wrapper colour, an unpackaged snack, and a handwritten note, with close agreement between what parents believed mattered and what educators reported checking. Over the audit window the sticker-award rate rose from 38% to 81% of lunchboxes, while the independent rubric's own nutrition score held within a narrow band throughout. A mid-study revision of the sticker criteria, intended to weight less visually obvious features more heavily, produced no measurable change in families' time to relearn the formula. We report the interview and audit findings together, note the limits of a six-centre sample, and reflect on what a check performed once, briefly, at a drop-off desk is actually able to ask of a family.

### Introduction

Slop University's evaluation-ecosystem agenda has spent several years cataloguing the distance between what an institutional check certifies and what it can actually see in the seconds it is given to look. Most of that catalogue concerns adults: sign-in sheets, verification prompts, indicator dashboards [1]. The daily healthy-pack sticker awarded at a childcare centre's drop-off desk is among the earliest checks most families ever encounter, and it shares the same structure as its adult-facing cousins: a single, quick, largely visual judgement, standing in for something — a lunchbox's actual nutritional balance — that would take considerably longer to assess.

Centres vary in how the sticker criteria are documented, and in our six sites none of them were written down in a form families could consult; the criteria existed only as what a room educator had learned, over some number of drop-offs, to

look for at a glance. This paper asks two related questions. First, what do families and educators actually believe the sticker is checking, and how closely do the two groups agree? Second, once a family has learned whatever the check rewards, how quickly does the lunchbox's real composition diverge from it, and does making the criteria harder to read at a glance change anything about how fast families relearn them?

We make three contributions, each hedged appropriately for a six-centre sample:

- We identify, via reflexive thematic analysis of interviews with parents and room educators, a small and largely agreed-upon set of visual cues the sticker appears to reward, none of which map cleanly onto a lunchbox's fruit, vegetable, or discretionary-food content.
- We track, via a fourteen-week photographic audit scored against a nutrition rubric kept independent of and blind to the sticker outcome,

a sticker-award rate that rises steadily as the cues become known while the rubric's own score does not follow it.

- We report a natural mid-study revision of the sticker criteria and find it changes, at most, how long a family takes to notice what changed, rather than whether the gap between the check and what it certifies closes.

## Related work

Measuring what actually ends up in a packed lunch is its own small literature. The School Food Checklist established that a simple, structured photograph-based audit tool can score a packed lunch's food categories reliably against direct observation [2], and a broader systematic review catalogues the tools — menu review, observation, weighed protocol, photography — available for the job, none of them cheap enough for a centre to run every day [3]. Photographic scoring of parent-packed lunches has since been used directly to characterise what preschoolers are actually sent with, alongside parent interviews about the trade-offs behind those choices [4], and a South Australian audit of school lunchboxes extends the same photographic method to packaging and food waste [5]. The Cafeteria Assessment for Elementary Schools takes a parallel approach at the point of consumption rather than packing, scoring a cafeteria's own physical environment against what students actually eat [6].

A second literature intervenes directly on what gets packed. Randomised and quasi-experimental programs targeting parents directly — Munch and Move [7], Lunch is in the Bag [8], SWAP IT [9], and an mHealth nudge delivered through an early-childhood-education app [10] — report modest, uneven gains in the discretionary-food share of a packed lunch, rarely sustained without ongoing contact; provider-facing capacity-building models such as Appetite to Play instead target the confidence and practice of the educators making mealtime decisions, one level removed from the lunchbox itself [11]. Centre-level healthy-eating practice more broadly predicts what children actually eat while in care, independent of any single named program [12]. None of this literature asks what a family or educator believes a same-day visual check is rewarding, which is the gap this paper works into — an omission consistent with a wider pattern in which

a quick institutional proxy outruns scrutiny of what it is actually measuring [13].

## Method

We recruited 34 parents and 11 room educators across six childcare centres in a single metropolitan network, purposively sampled to include at least one centre that had operated its healthy-pack sticker for under six months and at least one that had run it for more than three years [14]. Semi-structured interviews (20-35 minutes) asked educators what they looked for when deciding whether a lunchbox earned the sticker, and asked parents what they believed the sticker rewarded and how, if at all, that belief had changed their packing. Interviews continued at each site until two consecutive interviews produced no new codes, consistent with reported saturation thresholds for a moderately homogeneous sample [15]. Transcripts were coded independently by two authors following a reflexive thematic analysis procedure [16]; codebook disagreements were resolved by discussion, and the resulting cue-level categorisation (see Results) reached an inter-rater reliability of  $\kappa = 0.81$  on a 20% double-coded subsample [17].

In parallel, a research assistant blind to the interview findings photographed lunchboxes at drop-off across the same six centres for fourteen consecutive weeks ( $n = 1842$  lunchboxes), recording only whether the centre's own sticker was awarded that day. Each photograph was independently scored against a fixed nutrition rubric developed for this audit and never shared with centre staff, in which a lunchbox's fruit, vegetable, protein, and whole-grain content each contribute positively and its discretionary-snack share negatively:

$$R = w_1F + w_2V + w_3P + w_4G - w_5D$$

with weights  $w_i$  fixed before scoring began and set in inverse proportion to how visible each category is on an unopened glance — vegetables and protein, easily hidden beneath a wrapper, are weighted most heavily; a visible whole fruit least. At week eight, the network's own review process (not the research team) revised the sticker criteria to formally require a specific proportion of the lunchbox's visible area to show unwrapped produce, in response to informal educator concern that the sticker was “too easy to fake.” We treat this revision as a natural quasi-

experiment, comparing the days each family took to resume earning the sticker after the change against a matched pre-revision baseline (a cue change reported in an earlier term at two of the six centres).

## Results

Coding converged on four visual cues educators reported using to award the sticker at a glance: a whole piece of fruit visible on top of the box, bright wrapper colour, an unpackaged snack, and a handwritten note from home.

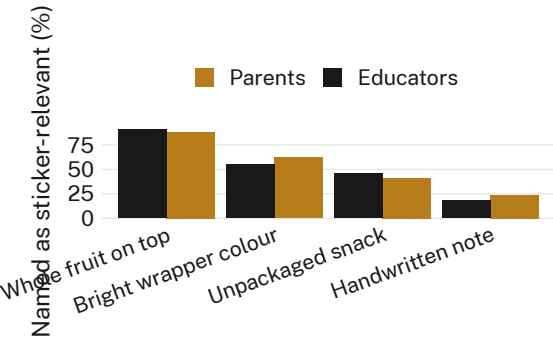


Figure 1: Parents and educators name almost the same cues as sticker-relevant, converging most on visible whole fruit (88% vs 91%) and least on an unpackaged snack (41% vs 46%).

No parent or educator, across either group, named vegetable content, protein content, or overall balance as something the sticker checked directly.

Across the fourteen-week audit, the sticker-award rate rose steadily from 38% of lunchboxes in week 1 to 81% in week 14, tracking closely with how long each centre had run its healthy-pack sticker before the audit began.

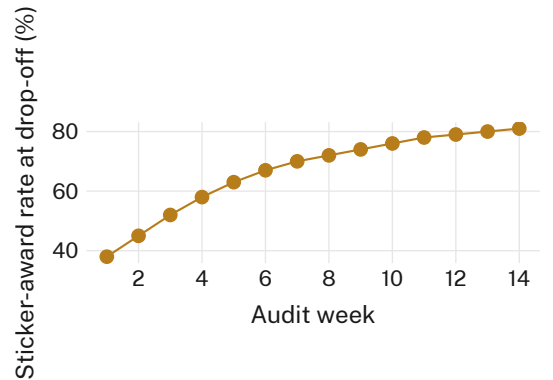


Figure 2: The sticker-award rate climbs from 38% to 81% across fourteen weeks of the photographic audit.

The independent nutrition rubric, scored blind to sticker outcome, told a flatter story: its mean score held between 5.1 and 5.6 out of a possible 10 across the same fourteen weeks (Table 1), with no week differing from the fourteen-week mean by more than 0.4 points. The rising sticker-award rate and the flat rubric score describe the same 1,842 lunchboxes.

| Audit week | Sticker-award rate (%) | Rubric score (mean, /10) |
|------------|------------------------|--------------------------|
| 1          | 38                     | 5.4                      |
| 4          | 58                     | 5.2                      |
| 8          | 72                     | 5.6                      |
| 11         | 78                     | 5.3                      |
| 14         | 81                     | 5.5                      |

Table 1: Rubric score by audit week, alongside the same weeks' sticker-award rate.

The week-8 rubric revision, intended by the centres' own review process to reward a genuine proportion of visible produce rather than a single piece of fruit, produced no reliable change in how quickly families relearned the criteria.

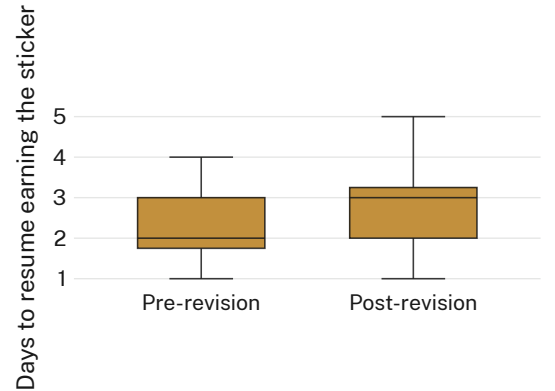


Figure 3: Days to resume earning the sticker after a cue change, before and after the week-8 revision: the two distributions barely separate (Mann-Whitney *U*, n.s.).

Families took a median of 2 days to resume earning the sticker after the earlier, informal cue changes at the two centres where we could establish a pre-revision baseline, against a median of 3 days after the formal week-8 revision; we retain the comparison for completeness.

## Discussion

Our results suggest a lunchbox sticker checked once, at a glance, at drop-off mainly rewards

a small, stable, and mutually-understood set of surface cues rather than any dimension of the lunch's actual nutritional balance, a pattern broadly consistent with the wider literature on quick institutional proxies drifting from what they were meant to certify [13]. That the network's own attempt to correct the sticker's criteria left the underlying gap unchanged is, we think, the more interesting finding: a revision aimed at the visual formula changed only how long relearning that formula took families, not whether relearning the formula remained the actual task at hand. We observe this pattern across six centres in one metropolitan network, and while the consistency across sites of quite different sticker tenure suggests the finding may generalise beyond any one centre's particular educator or wording, we make that suggestion cautiously.

## Limitations

Our sample of six centres and 1,842 audited lunchboxes is modest relative to the multi-site randomised evaluations reviewed above [7], [9], though the range of sticker tenure it captures — from under six months to over three years — is broader than most single-program evaluations attempt. The pre-revision baseline for the days-to-relearn comparison draws on only two of the six centres, a constraint imposed by which sites had a documented earlier cue change to compare against; the two sites nonetheless bracket the network's youngest and oldest sticker programs. We did not weigh or directly assay lunchbox contents, relying instead on a validated photographic scoring approach [2], [4]; the resulting rubric score is consequently a proxy in its own right, a limitation we consider in keeping with the paper's subject.

## Conclusion

A daily healthy-pack sticker, checked once and briefly at a childcare centre's drop-off desk, rewards what a glance can see: a whole fruit on top, a bright wrapper, an unpackaged snack, a note from home. Families and educators agree closely on what those cues are, learn them within weeks, and relearn them within days of any change — including a change the network's own review process designed specifically to be harder to game [18]. Whether a daily visual check can ever certify more than what it can see in the time it is given remains, on this evi-

dence, an open question; future work might test whether a slower, less frequent check — one that trades the sticker's immediacy for something closer to the rubric behind it — changes what families actually pack, rather than only what they show at the door.

## References

- [1] D. Okafor, B. Ntuli, M. Hanke, and R. Sabel, "Signed, Not Secured: Auditing a Term of Paper Sign-In Sheets After Improvement Grand Rounds Retired Its Tap-Card System," *Slop University technical report*, 2026, doi: [10.5555/slop.k7y0kk](https://doi.org/10.5555/slop.k7y0kk).
- [2] S. A. Mitchell, C. L. Miles, L. Brennan, and J. Matthews, "Reliability of the School Food Checklist for in-school audits and photograph analysis of children's packed lunches," *Journal of Human Nutrition and Dietetics*, vol. 23, no. 1, pp. 48–53, 2010, doi: [10.1111/j.1365-277X.2009.00996.x](https://doi.org/10.1111/j.1365-277X.2009.00996.x).
- [3] A. Elford et al., "Identification and Evaluation of Tools Utilised for Measuring Food Provision in Childcare Centres and Primary Schools: A Systematic Review," *International Journal of Environmental Research and Public Health*, vol. 19, no. 7, p. 4096, 2022, doi: [10.3390/ijerph19074096](https://doi.org/10.3390/ijerph19074096).
- [4] S. Elliott, M. L. McCloskey, S. L. Johnson, N. Z. Mena, T. Swindle, and L. L. Bellows, "Food Photography as a Tool to Assess Type, Quantity, and Quality of Foods in Parent-Packed Lunches for Preschoolers," *Journal of Nutrition Education and Behavior*, vol. 53, no. 2, pp. 164–173, 2021, doi: [10.1016/j.jneb.2020.10.007](https://doi.org/10.1016/j.jneb.2020.10.007).
- [5] N. K. Lalchandani, C. Hume, L. Giles, S. Crabb, J. Hendrikx, and C. Miller, "Sneak peek: food, waste and packaging characteristics of South Australian school children's lunchboxes," *Public Health Nutrition*, vol. 28, no. 1, 2025, doi: [10.1017/S1368980025000126](https://doi.org/10.1017/S1368980025000126).
- [6] K. A. Rollings and N. M. Wells, "Cafeteria assessment for elementary schools (CAFES): development, reliability testing, and predictive validity analysis," *BMC Public Health*, vol. 18, 2018, doi: [10.1186/s12889-018-6032-2](https://doi.org/10.1186/s12889-018-6032-2).
- [7] L. L. Hardy, L. King, B. Kelly, L. Farrell, and S. Howlett, "Munch and Move: evaluation of

- a preschool healthy eating and movement skill program,” *International Journal of Behavioral Nutrition and Physical Activity*, vol. 7, p. 80, 2010, doi: [10.1186/1479-5868-7-80](https://doi.org/10.1186/1479-5868-7-80).
- [8] C. Roberts-Gray *et al.*, “Efficacy of the Lunch is in the Bag intervention to increase parents' packing of healthy bag lunches for young children: a cluster-randomized trial in early care and education centers,” *International Journal of Behavioral Nutrition and Physical Activity*, vol. 13, 2016, doi: [10.1186/s12966-015-0326-x](https://doi.org/10.1186/s12966-015-0326-x).
- [9] R. Sutherland *et al.*, “A randomized controlled trial to assess the potential efficacy, feasibility and acceptability of an m-health intervention targeting parents of school aged children to improve the nutritional quality of foods packed in the lunchbox ‘SWAP IT’,” *International Journal of Behavioral Nutrition and Physical Activity*, vol. 16, 2019, doi: [10.1186/s12966-019-0812-7](https://doi.org/10.1186/s12966-019-0812-7).
- [10] N. Pearson *et al.*, “An mHealth Intervention to Reduce the Packing of Discretionary Foods in Children's Lunch Boxes in Early Childhood Education and Care Services: Cluster Randomized Controlled Trial,” *Journal of Medical Internet Research*, vol. 24, no. 3, p. e27760, 2022, doi: [10.2196/27760](https://doi.org/10.2196/27760).
- [11] E. J. Buckler *et al.*, “Scaling up healthy eating in early childhood education and care: evaluation of the Appetite to Play capacity-building intervention,” *Public Health Nutrition*, vol. 27, no. 1, 2024, doi: [10.1017/S1368980024001290](https://doi.org/10.1017/S1368980024001290).
- [12] C. Barnes *et al.*, “The Association between Australian Childcare Centre Healthy Eating Practices and Children's Healthy Eating Behaviours: A Cross-Sectional Study within Lunchbox Centres,” *Nutrients*, vol. 13, no. 4, p. 1139, 2021, doi: [10.3390/nu13041139](https://doi.org/10.3390/nu13041139).
- [13] G. Bevan and C. Hood, “What's Measured Is What Matters: Targets and Gaming in the English Public Health Care System,” *Public Administration*, vol. 84, no. 3, pp. 517–538, 2006, doi: [10.1111/j.1467-9299.2006.00600.x](https://doi.org/10.1111/j.1467-9299.2006.00600.x).
- [14] L. A. Palinkas, S. M. Horwitz, C. A. Green, J. P. Wisdom, N. Duan, and K. Hoagwood, “Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research,” *Administration and Policy in Mental Health and Mental Health Services Research*, vol. 42, no. 5, pp. 533–544, 2015, doi: [10.1007/s10488-013-0528-y](https://doi.org/10.1007/s10488-013-0528-y).
- [15] G. Guest, A. Bunce, and L. Johnson, “How Many Interviews Are Enough? An Experiment with Data Saturation and Variability,” *Field Methods*, vol. 18, no. 1, pp. 59–82, 2006, doi: [10.1177/1525822X05279903](https://doi.org/10.1177/1525822X05279903).
- [16] V. Braun and V. Clarke, “Using thematic analysis in psychology,” *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006, doi: [10.1191/1478088706qp063oa](https://doi.org/10.1191/1478088706qp063oa).
- [17] M. L. McHugh, “Interrater reliability: the kappa statistic,” *Biochemia Medica*, vol. 22, no. 3, pp. 276–282, 2012, doi: [10.11613/BM.2012.031](https://doi.org/10.11613/BM.2012.031).
- [18] V. Marris and K. Lindqvist, “Tagged to Stay Put: Forecasting Missing AV Trolleys Before the Weekly Stocktake,” *Slop University technical report*, 2026, doi: [10.5555/slop.mf9gpt](https://doi.org/10.5555/slop.mf9gpt).